

4. Given an $m \times n$ grid and a ball at a starting cell, find the number of ways to move the ball out of the grid boundary in exactly N steps.

Example:

· **Input:** $m=2, n=2, N=2, i=0, j=0$ · **Output:** 6

· **Input:** $m=1, n=3, N=3, i=0, j=1$ · **Output:** 12

Program:

```
#include <stdio.h>

int m,n,N;

int ways(int i,int j,int steps){
    if(i<0 || i>=m || j<0 || j>=n)
        return 1;
    if(steps==0)
        return 0;
    return ways(i-1,j,steps-1)+ways(i+1,j,steps-1)+ways(i,j-1,steps-1)+ways(i,j+1,steps-1);
}

int main(){
    int i,j;
    printf("Enter rows: ");
    scanf("%d",&m);
    printf("Enter columns: ");
    scanf("%d",&n);
    printf("Enter number of steps: ");
    scanf("%d",&N);
    printf("Enter starting row: ");
    scanf("%d",&i);
    printf("Enter starting column: ");
    scanf("%d",&j);
    printf("Number of ways: %d",ways(i,j,N));
    return 0;
}
```

Input & Output :

<pre> Enter rows: 2 Enter columns: 2 Enter number of steps: 2 Enter starting row: 0 Enter starting column: 0 Number of ways: 6 ...Program finished with exit code 0 Press ENTER to exit console. </pre>	<pre> Enter rows: 1 Enter columns: 3 Enter number of steps: 3 Enter starting row: 0 Enter starting column: 1 Number of ways: 12 ...Program finished with exit code 0 Press ENTER to exit console. </pre>
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5. You are a professional robber planning to rob houses along a street. Each house has a certain amount of money stashed. All houses at this place are arranged in a circle. That means the first house is the neighbor of the last one. Meanwhile, adjacent houses have security systems connected, and it will automatically contact the police if two adjacent houses were broken into on the same night.

Examples:

Input : nums = [2, 3, 2]

Output : The maximum money you can rob without alerting the police is 3(robbing house 1).

(ii) Input : nums = [1, 2, 3, 1]

Output : The maximum money you can rob without alerting the police is 4 (robbing house 1 and house 3).

Program :

```

#include <stdio.h>

int max(int a,int b){
    if(a>b)
        return a;
    return b;
}

int rob(int a[],int start,int end){
    int prev=0,curr=0,next,i;
    for(i=start;i<=end;i++){
        next=max(curr,prev+a[i]);
        prev=curr;
        curr=next;
    }
}

```

```

    }

    return curr;
}

int main(){
    int n,i,a[100],ans1,ans2;

    printf("Enter number of houses: ");

    scanf("%d",&n);

    printf("Enter money in each house: ");

    for(i=0;i<n;i++)

        scanf("%d",&a[i]);

    if(n==1){

        printf("Maximum money: %d",a[0]);

        return 0;

    }

    ans1=rob(a,0,n-2);

    ans2=rob(a,1,n-1);

    printf("Maximum money: %d",max(ans1,ans2));

    return 0;

}

```

Input & Output :

```

Enter number of houses: 3
Enter money in each house: 2 3 2
Maximum money: 3

...Program finished with exit code 0
Press ENTER to exit console.

```

```

Enter number of houses: 4
Enter money in each house: 1 2 3 1
Maximum money: 4

...Program finished with exit code 0
Press ENTER to exit console.

```

6. You are climbing a staircase. It takes n steps to reach the top. Each time you can either climb 1 or 2 steps. In how many distinct ways can you climb to the top?

Examples:

Input: $n=4$ Output: 5

Input: $n=3$ Output: 3

Program :

```
#include <stdio.h>

int main(){
    int n,i,a=1,b=1,c;

    printf("Enter number of steps: ");

    scanf("%d",&n);

    if(n==0){
        printf("Number of ways: 1");

        return 0;
    }

    for(i=2;i<=n;i++){
        c=a+b;

        a=b;

        b=c;
    }

    printf("Number of ways: %d",b);

    return 0;
}
```

Input & Output :

<pre>Enter number of steps: 4 Number of ways: 5 ...Program finished with exit code 0 Press ENTER to exit console.</pre>	<pre>Enter number of steps: 3 Number of ways: 3 ...Program finished with exit code 0 Press ENTER to exit console.</pre>
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7. A robot is located at the top-left corner of a $m \times n$ grid .The robot can only move either down or right at any point in time. The robot is trying to reach the bottom-right corner of the grid. How many possible unique paths are there?

Examples:

Input: $m=7,n=3$ **Output:** 28

Input: $m=3,n=2$ **Output:** 3

Program :

```
#include <stdio.h>
```

```

int main(){

    int m,n,i,j;

    int a[100][100];

    printf("Enter rows: ");

    scanf("%d",&m);

    printf("Enter columns: ");

    scanf("%d",&n);

    for(i=0;i<m;i++){

        for(j=0;j<n;j++){

            if(i==0 || j==0)

                a[i][j]=1;

            else

                a[i][j]=a[i-1][j]+a[i][j-1];

        }

    }

    printf("Number of unique paths: %d",a[m-1][n-1]);

    return 0;

}

```

Input & Output :

```

Enter rows: 7
Enter columns: 3
Number of unique paths: 28

...Program finished with exit code 0
Press ENTER to exit console.

```

```

Enter rows: 3
Enter columns: 2
Number of unique paths: 3

...Program finished with exit code 0
Press ENTER to exit console.

```